

Question	Answer	Marks
3(a)(i)	Using $I = \frac{P}{4\pi r^2}$	1
	$I = \frac{3.826 \times 10^{26}}{4\pi(149.6 \times 10^6 \times 10^3)^2} = 1360.4 \text{ W m}^{-2}$ (at least 4 s.f)	1
(a)(ii)	(Any 1 point) - Sun emits light across entire EM spectrum, but the photometer in question detects only visible region - Atmosphere absorbs or reflects some of EM radiation. - Presence of clouds block some of the sunlight. - Angle at which sun's rays strike earth's surface not right angles.	1
(a)(iii)	$I = \frac{P}{A}$	1
	$P_{max} = I_{max} \cdot A = 1360.4(0.0400) = 54.42 \text{ W}$ (3 s.f)	1
(b)(i)	$y = y_o \cos \omega t$	1
	$y = 0.020 \cos\left(\frac{2\pi}{16}\right)t = 0.020 \cos 0.125\pi t = 0.020 \cos 0.393t$	1
	1 mark for correct ω	
(b)(ii)	$\Delta\phi = \frac{x}{\lambda} \times 2\pi = \frac{\lambda/4}{\lambda} \times 2\pi$	1
	$= 0.50 \pi \text{ rad} = 1.57 \text{ rad}$ (no units deduct one mark)	1
(b)(iii)	- negative sine graph (correct shape) - with same period of 16 s - smooth and sinusoidal curve (2 marks) Mark by deduction	
	Max mark	11

Question	Answer	Marks
4(a)	A region of space where a charged particle will experience an electric force	1